ELEC50011A - MATHEMATICS 2A-2026-SOLUTIONS

1. a) Given the function $u(x,y) = -e^{-y}\sin x - e^{-x}\cos y$,

(i) show that $u(x,y)$ satisfies Laplace's equation and integrate the Cauchy-Riemann equations to find its harmonic conjugate $v(x,y)$; [5]

SOLUTION

First differentiate:

$u_x = -e^{-y}\cos x + e^{-x}\cos y \Rightarrow u_{xx} = e^{-y}\sin x - e^{-x}\cos y$ and

$u_y = e^{-y}\sin x + e^{-x}\sin y \Rightarrow u_{yy} = -e^{-y}\sin x + e^{-x}\cos y$

and hence $u_{xx} + u_{yy} = 0$, so u is a harmonic function, satisfying Laplace's equation. Therefore u has a harmonic conjugate v and together they satisfy the Cauchy-Riemann equations: $u_x = v_y$ and $u_y = -v_x$. Hence

$$v = \int u_x\,dy = \int -e^{-y}\cos x + e^{-x}\cos y\,dy = e^{-y}\cos x + e^{-x}\sin y + A(x),$$

and

$$v = \int -u_y\,dx = -\int e^{-y}\sin x + e^{-x}\sin y\,dx = e^{-y}\cos x + e^{-x}\sin y + B(y),$$

where A and B are arbitrary functions. Comparing the two, we see that $A = B =$constant, and

$$v(x,y) = e^{-y}\cos x + e^{-x}\sin y + C$$

is the harmonic conjugate, where $C \in \mathbb{R}$.

Mostly done well. A lof ot people forgot about $A(x)$ and $B(y)$.

(ii) hence obtain $f(z) = u(x,y) + iv(x,y)$ where f is an analytic function of $z = x + iy$, simplifying as much as possible. [5]

SOLUTION

Joining the two, we have

$$\begin{aligned} f(z) &= u + iv = -e^{-y}\sin x - e^{-x}\cos y + i(e^{-y}\cos x + e^{-x}\sin y + C) \\ &= e^{-y}(i\cos x - \sin x) - e^{-x}(\cos y - i\sin y) + Ci \\ &= ie^{-y}(\cos x + i\sin x) - e^{-x}e^{-iy} + Ci = ie^{-y}e^{ix} - e^{-x-iy} + Ci \\ &= ie^{iz} - e^{-z} + Ci \end{aligned}$$

Many incorrect versions here, mostly related to a misinterpretation of Euler's formula.

b) Let a mapping from the $z-$plane to the $w-$ plane be given by

$$w(z) = \frac{z}{i-z}.$$

Show that a family of circles C_1 in the $z-$plane, with centre $z = i$ and radius r, maps to a family of circles C_2 in the $w-$plane, stating their centre and radius.

Explain carefully what happens to points inside and points outside C_1 under the mapping. Finally, state the images under the map of the vertical axis, and the line $y = 1$. [10]

SOLUTION Begin with

$$w = -\frac{z}{z-i} = -\frac{z-i+i}{z-i} = -1-\frac{i}{z-i} \Rightarrow w+1 = -\frac{i}{[x+i(y-1)]}\frac{[x-i(y-1)]}{[x-i(y-1)]}$$

so

$$w+1 \;=\; u+1+iv \;=\; \frac{1-y-ix}{x^2+(1-y)^2}\,,$$

so we have

$$u+1 \;=\; \frac{1-y}{x^2+(1-y)^2} \quad \text{and} \quad v = -\frac{x}{x^2+(1-y)^2}$$

The family of circles with centre $z = i$, i.e. the point $(0,1)$, and radius r have equation $x^2+(y-1)^2 = r^2$. Hence we set

$$(u+1)^2+v^2 \;=\; \frac{(1-y)^2+x^2}{[x^2+(y-1)^2]^2} \;=\; \frac{1}{x^2+(y-1)^2} = \frac{1}{r^2}$$

for the given family of circles in the $z-$plane. Hence, these map to a family of circles in the $w-$plane, centred at $(-1,0)$, i.e $w = -1$ with radius $1/r$.

Probably the most difficult question on the exam. Most people tried calculating u^2+v^2 directly and many of these got lost trying to simplify. A minority went solved for z:

$$z = \frac{wi}{w+1} \Rightarrow z-i = -\frac{i}{w+1} \Rightarrow \left|-\frac{i}{w+1}\right| = r \Rightarrow |w+1| = \frac{1}{r}$$

Not everyone who took this approcach got to the correct C_2, but quite a few.

Points inside a circle in C_1 map to the outside of the corresponding circle in C_2 and vice-versa. This can be argued in various ways, but most easily using continuity.

Let $r = 2$. The circle in C_1 with equation $x^2+(y-1)^2 = 4$ has the origin as an interior point, and maps to the circle in C_2 with equation $(u+1)^2+v^2 = 1/4$, centred at $(-1,0)$, with radius $1/r = 1/2$. The

origin $z=0$ maps to the origin $w=0$, but this is outside the circle $(u+1)^2+v^2=1/4$. By continuity, this applies to every point inside a circle in C_1, and the same argument can be applied to a point on the outside.

Almost everyone attempted this, but the rigour of the arguments varied.

The vertical, $y-$axis has $x=0$, so maps to $v=0$, the horizontal axis. The line $y=1$ maps to $u=0$. the vertical axis.

Oddly, a lot of people who couldn't do the first part, just didn't attempt this, though it's the easiest part of the question.

2. a) Given the real integral

$$I=\int_0^{2\pi}\frac{d\theta}{(5+4\sin\theta)^2},$$

(i) Taking complex z, where $|z|=1$, show that we can rewrite I as a contour integral

$$I=\frac{1}{a}i\oint_C\frac{z\,dz}{(z+bi)^2\left(z+\frac{1}{b}i\right)^2},$$

giving the closed contour C, and integers a,b. . [5]

SOLUTION

With $|z|=1$, we know that C is the unit circle, where $z=e^{i\theta}$ gives the necessary substitution, with $\theta=0\ldots2\pi$; hence $d\theta=\dfrac{dz}{iz}$ and we use $\sin\theta=\frac{1}{2i}\left(z-\frac{1}{z}\right)$ to get

$$\begin{aligned}I&=\oint_C\frac{1}{\left[5+\frac{4}{2i}\left(z-\frac{1}{z}\right)\right]^2}\frac{dz}{iz}=-i\oint_C\frac{z\,dz}{\left[5z-2i\left(z^2-1\right)\right]^2}\\&=-\frac{1}{(-2i)^2}i\oint_C\frac{z\,dz}{\left(z^2+\frac{5}{2}iz-1\right)^2}\end{aligned}$$

Hence we have integer $a=4$, and solve

$$z^2+\frac{5}{2}iz-1=0\Rightarrow z=-\frac{5}{4}i\pm\frac{3}{4}i=-\frac{1}{2}i,-2i$$

giving the second integer $b=2$, and the integral becomes

$$I=\frac{1}{4}i\oint_C\frac{z\,dz}{(z+2i)^2\left(z+\frac{1}{2}i\right)^2},$$

as required.

Very many multiplied out $(5+4\sin\theta)^2$ *which then made the substitution with* $z-1/z$ *give a degree-4 polynomial. Of these, only a few managed the factorization correctly, often by multiplying out* $(z+bi)^2\left(z+\frac{1}{b}i\right)^2$ *and comparing coefficients.*

(ii) Using Cauchy's residue theorem, or otherwise, calculate I. [5]

Recall that the residue of a complex function $F(z)$ *at a pole* $z=a$ *of multiplicity* m *is given by the expression*

$$\lim_{z\to a}\left\{\frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}\left[(z-a)^m F(z)\right]\right\}.$$

SOLUTION

To use the residue theorem, we need to calculate the residues due to each pole inside C, and there is only the double pole at $z=-\frac{1}{2}i$, as $z=-2i$ is outside the unit circle. Using the residue formula we obtain:

$$\lim_{z\to-\frac{1}{2}i}\frac{d}{dz}\left[\left(z+\frac{1}{2}i\right)^2\left(\frac{z}{(z+2i)^2\left(z+\frac{1}{2}i\right)^2}\right)\right]=\lim_{z\to-\frac{1}{2}i}\frac{d}{dz}\left(\frac{z}{(z+2i)^2}\right)$$

$$=\lim_{z\to-\frac{1}{2}i}\frac{-z+2i}{(z+2i)^3}\;=\;\frac{\frac{5}{2}i}{\left(\frac{3}{2}i\right)^3}=-\frac{20}{27},$$

after some simplification. Cauchy's residue theorem gives that

$$I=\frac{1}{4}i\times 2\pi i\times(\sum\text{ Residues inside }C)=-\frac{1}{2}\pi\times\left(-\frac{20}{27}\right)\;=\;\frac{10\pi}{27}.$$

A lot of arithmetic error here, when finding the residue: quotient rule, powers of i, etc. A lot of people calculated both residues, sometimes correctly, and then put both into Cauchy's Residue Theorem.

b) Prove that real, symmetric matrices have real eigenvalues. [6]

Solution: We begin with the eigenvalue equation and take complex conjugates

$$A\underline{\mathbf{x}}=\lambda\underline{\mathbf{x}}\quad\Rightarrow\quad\bar{A}\bar{\underline{\mathbf{x}}}=\bar{\lambda}\bar{\underline{\mathbf{x}}}\quad\Rightarrow\quad A\bar{\underline{\mathbf{x}}}=\bar{\lambda}\bar{\underline{\mathbf{x}}},$$

as A has real entries. Now take the transpose:

$$(A\bar{\underline{\mathbf{x}}})^T=(\bar{\lambda}\bar{\underline{\mathbf{x}}})^T\Rightarrow\bar{\underline{\mathbf{x}}}^TA^T=\bar{\underline{\mathbf{x}}}^T\bar{\lambda}\Rightarrow\bar{\underline{\mathbf{x}}}^TA=\bar{\underline{\mathbf{x}}}^T\bar{\lambda}$$

as A is symmetric. Multiply the last equation on the right by $\underline{\mathbf{x}}$ to get

$$(\bar{\underline{\mathbf{x}}}^TA)\underline{\mathbf{x}}=(\bar{\underline{\mathbf{x}}}^T\bar{\lambda})\underline{\mathbf{x}}=\bar{\lambda}\,|\underline{\mathbf{x}}|^2\qquad(1)$$

and multiply the eigenvalue equation on the left by $\bar{\underline{\mathbf{x}}}^T$

$$\bar{\underline{\mathbf{x}}}^T(A\underline{\mathbf{x}}) = \bar{\underline{\mathbf{x}}}^T(\lambda\underline{\mathbf{x}}) \;=\; \lambda|\underline{\mathbf{x}}|^2 \qquad (2)$$

Matrix multiplication is associative, so the lhs of both (1) and (2) are the same, so the rhs is also equal:

$$\lambda|\underline{\mathbf{x}}|^2 \;=\; \bar{\lambda}|\underline{\mathbf{x}}|^2 \Rightarrow \lambda \;=\; \bar{\lambda}$$

as $|\underline{\mathbf{x}}| \neq 0$, given that it's an eigenvector. So $\lambda \;=\; \bar{\lambda}$, and is therefore real.

Many found this difficult, quite a few tried to prov that eigenvectors corresponding to distinct eigenvalues are orthogonal. Where the proof was done mostly correctly, the most common error, surprisingly, was people not mentioning that in the last step the cancellation works due to $|\underline{\mathbf{x}}| \neq 0$*, though this is crucial.*

c) Obtain the orthogonal diagonalization of the symmetric matrix

$$A = \begin{pmatrix} 9 & -2 \\ -2 & 6 \end{pmatrix} . \qquad [\,6\,]$$

Solution: We find the eigenvalues solving

$$\det(A-\lambda I) = 0 \Rightarrow (9-\lambda)(6-\lambda) - 4 = 0 \Rightarrow \lambda = 5, 10\,.$$

The normalized eigenvectors are found for $\lambda = 5$:

$$(A-5I)\underline{\mathbf{x}}_1 = \underline{\mathbf{0}} \Rightarrow \begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix}\underline{\mathbf{x}}_1 = \underline{\mathbf{0}} \Rightarrow \underline{\mathbf{x}}_1 = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

and for $\lambda = 10$:

$$(A-10I)\underline{\mathbf{x}}_2 = \underline{\mathbf{0}} \Rightarrow \begin{pmatrix} -1 & -2 \\ -2 & -4 \end{pmatrix}\underline{\mathbf{x}}_2 = \underline{\mathbf{0}} \Rightarrow \underline{\mathbf{x}}_2 = \frac{1}{\sqrt{5}}\begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

The eignenvectors are orthonormal, and form the columns of the orthogonal matrix S, and we have

$$S \;=\; \frac{1}{\sqrt{5}}\begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} \Rightarrow A \;=\; S\Lambda S^T\,,$$

the orthogonal diagonalization of A, where $\Lambda \;=\; \begin{pmatrix} 5 & 0 \\ 0 & 10 \end{pmatrix}$.

The best question, in terms of number of correct answers. A few arithmetic errors. Some people didn't normalize the eigenvectors.

d) Given

$$A = \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \end{pmatrix}\,,$$

its Singular Value Decomposition (SVD) is $A = U\Sigma V^T$. Obtain U, V and Σ, stating which bases of fundamental subspaces are involved, and use these to write A as a sum of rank-1 matrices. [8]

Solution: As A is 2×3 we require the smaller matrix AA^T which will be 2×2:

$$AA^T \;=\; B \;=\; \begin{pmatrix} 9 & -2 \\ -2 & 6 \end{pmatrix}.$$

This is the matrix from the previous question, so we already have eigenvectors

$$\underline{\mathbf{u}}_1 \;=\; \frac{1}{\sqrt{5}}\begin{pmatrix} -2 \\ 1 \end{pmatrix} \quad \text{and} \quad \underline{\mathbf{u}}_2 \;=\frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

with respective eigenvalues $\lambda = 10, 5$. These form a basis of the column space $C(A)$, and we have the first orthogonal matrix

$$U \;=\; \frac{1}{\sqrt{5}}\begin{pmatrix} -2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The singular values are the roots of the eigenvalues so

$$\Sigma \;=\; \begin{pmatrix} \sqrt{10} & 0 & 0 \\ 0 & \sqrt{5} & 0 \end{pmatrix}$$

To find $\underline{\mathbf{v}}_i$ we use $\underline{\mathbf{v}}_i = \frac{1}{\sigma_i}A^T\underline{\mathbf{u}}_i$ and get

$$\underline{\mathbf{v}}_1 \;=\; \frac{1}{\sqrt{10}}\begin{pmatrix} 1 & 2 \\ -2 & 1 \\ 2 & -1 \end{pmatrix}\frac{1}{\sqrt{5}}\begin{pmatrix} -2 \\ 1 \end{pmatrix} \;=\; \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix},$$

and

$$\underline{\mathbf{v}}_2 \;=\; \frac{1}{\sqrt{5}}\begin{pmatrix} 1 & 2 \\ -2 & 1 \\ 2 & -1 \end{pmatrix}\frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \end{pmatrix} \;=\; \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

These two vectors are a basis for $C(A^T)$, the row space of A. Finally, the third column of V is a basis for $N(A)$, the nullspace of A and orthogonal complement of the row space, so we solve

$$A\underline{\mathbf{x}} = \underline{\mathbf{0}} \Rightarrow \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \end{pmatrix}\underline{\mathbf{x}} = \underline{\mathbf{0}}\,.$$

A single row operation gives $z = y$ and $x = 0$, so after normalization, we have

$$\underline{\mathbf{v}}_3 \;=\; \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \;\Rightarrow\; V \;=\; \frac{1}{\sqrt{2}}\begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix}$$

We can use the SVD to write A as a sum of rank-1 matrices, taking

$$A \;=\; \sigma_1\underline{\mathbf{u}}_1\underline{\mathbf{v}}_1^T + \ldots + \sigma_r\underline{\mathbf{u}}_r\underline{\mathbf{v}}_r^T$$

and here $r = 2$, so

$$A = \sigma_1\underline{\mathbf{u}}_1\underline{\mathbf{v}}_1^T + \sigma_2\underline{\mathbf{u}}_2\underline{\mathbf{v}}_2^T = \sqrt{10}\frac{1}{\sqrt{5}}\begin{pmatrix} -2 \\ 1 \end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & -1 \end{pmatrix} + \sqrt{5}\frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -2 & 2 \\ 0 & 1 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}.$$

A lot of good answers. The most common error was in finding the sum of rank-1 matrices: a lot of people didn't remember this. Other common errors included wrong matrix size for Σ, forgetting $N(A)$ and writing V as 3×2, the order of eigenvectors in U and/or V not matching the order of eigenvalues in Σ. A substantial number of people who had U correctly, then calculated the eigenvalues and eigenvectors of A^TA to find $\underline{\mathbf{v}}_{1,2}$ but couldn't work out that the zero eigenvalue gives the third eigenvector. Many people began with A^TA and had a lot more work, as they didn't see the connection to question 2c.